

Data Visualisation

Waiting for students to log in

Data Visualisation

Lecture 1

Preliminaries

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Brightspace

Data Visualisation SPEC9995: 2026-2027

Signing sheet

Credits

This course is based on material developed by :

Marisa Llorens-Salvador

Cathy Ennis, John McAuley, Brian Mac Namee and Colman McMahon

Material from other sources is also used (citations included)

Course details

- Lectures: Online (recorded)
 - Tuesday 18:30 -20:00
- Weekly Labs (10%): 5 of these labs are worth 2% each
 - After Tuesday lectures –online 20:00-21:30
- Assignment 1 (30%): Visualisation project using Tableau
- Assignment 2 (30%): Project using the statistics package R
- Exam (30%)

Course Overview

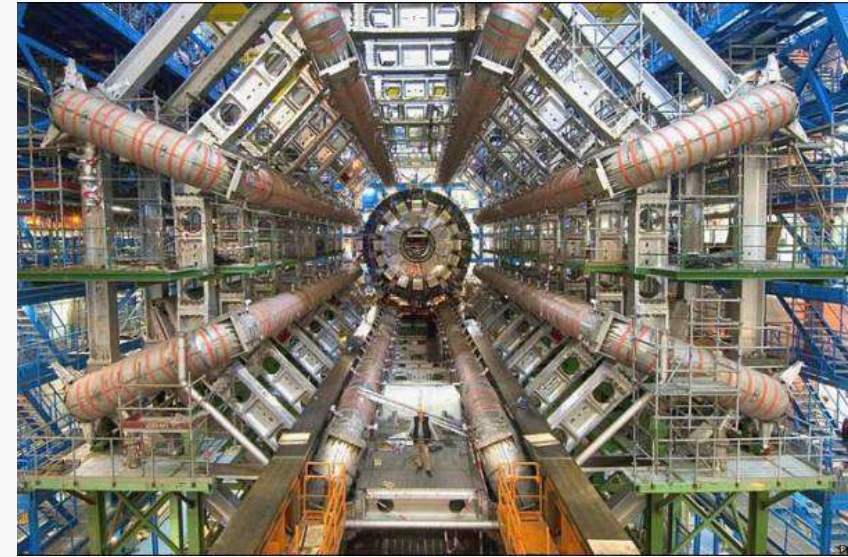
Week	Lecture topic	Lab Topic	Lab Marks	Assignment Due date
1	Introduction	Visualisation discussion		
2	Perception/Classification	Intro to Tableau		
3	Encodings 1	Tableau Visualisation	2%	
4	Encodings 2	Dashboards	2%	
5	Visualising trends	Exploratory Analysis	2%	
6	Visualising Comparisons	Catch up		
7	Accessibility in Visualisation	Intro to R*		30%Tableau
8				
9	Visualising Relationships	Visualisation in R		
10	Visualising Text	Visualisation in R	2%	
11	Visualising geospatial data	Visualising geospatial data	2%	
12	Visualising Spatial Relationships	Catch up and marking		
13	Review week			30% R

Visualisation

Drowning in Data

- The Large Hadron Collider at CERN is the largest scientific instrument ever constructed
- When turned on, Particles collide in the LHC detectors approximately 1 billion times per second, generating about one petabyte of collision data per second
- Storing only 'interesting' data keeps one petabyte a day

Petabyte = 1m Gb



15 minutes of Fame 2019-2020

- 500 hours per minute = 576 000 hours per day
- 5+ billion videos watched per day
- 5 billion videos shared to date
- 122 million users daily
- Most viewed video over 13 billion views
- First 24 hours 63 million views (Jan 2023)

<https://expandedramblings.com/index.php/youtube-statistics/>
<https://www.omnicoreagency.com/youtube-statistics/>



Example: Taxi!

I order a taxi from the airport...

How can this journey be logged?



Example: Taxi!

- The taxi operator can record the time the taxi was ordered, the pick-up and destination locations, the customer's details...
- The phone company can record the phone call maker and receiver, the duration of the call, ...
- The credit card processor can record the transaction amount, transaction time, transaction type, ...
- The credit card issuer can record the billing record, the interest rate, the current available credit ...
- The car maintenance service can record the fuel usage, the engine state ...
- The passenger can record the expense of the taxi
- The DAA can record the time the taxi drop off, the number of passengers in the car...
- Dublin City Council can record the passage of the taxi across the city...
- The taxi driver can record the time of the fare, the type of fare, the amount of the fare...
- The taxi company can record the path taken by the driver, the duration of the trip...

Questions?

1. How can we understand this data?
2. How can we learn for this data?
3. How can we make this data accessible?
4. How can we gain *insights* into this data?

What is Visualisation?

1. The action or fact of visualizing; the power or process of forming a mental picture or vision of something not actually present to the sight; a picture thus formed.
2. The action or process of rendering visible.
 - ✓ “Transformation of the symbolic into the geometric”
[McCormick et al. 1987]
 - ✓ “... finding the artificial memory that best supports our natural means of perception.”
[Bertin1967]
 - ✓ “The use of computer-generated, interactive, visual representations of data to amplify cognition.”
[Card, Mackinlay, & Shneiderman1999]

Why Visualisation?

- Answer questions (or discover them)
- Make decisions
- See data in context
- Expand memory
- Support graphical calculation
- Find patterns
- Present argument or tell a story
- Inspire

Why Visualisation?

Well-designed visual representations can replace cognitive calculations with simple perceptual inferences and improve comprehension, memory, and decision making

By making data more accessible and appealing, visual representations may also help engage more diverse audiences in exploration and analysis

The challenge is to create effective and engaging visualizations that are appropriate to the data

Functions of Visualisation

- **Record:** store information
 - Photographs, blueprints, ...
- **Analyze:** support reasoning about information
 - Process and calculate
 - Reason about data
 - Feedback and interaction
- **Communicate:** convey information to others
 - Share and persuade
 - Collaborate and revise
 - Emphasize important aspects of data

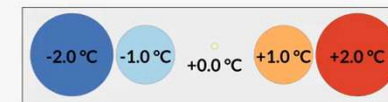
Trends over time



Trends over time

Temperature Anomalies by Country
Years 1880 - 2017

1880



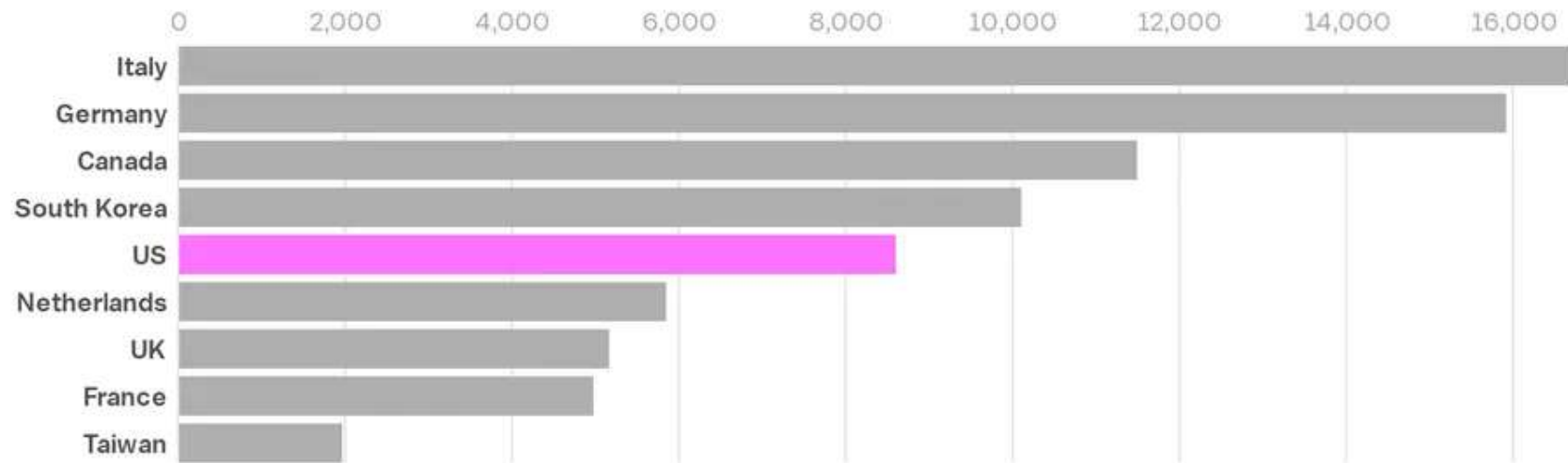
Data Source:
NASA GISS, GISTEMP Land-Ocean Temperature Index (LOTI), ERSSTv5, 1200km smoothing
<https://data.giss.nasa.gov/gistemp/>
Average of monthly temperature anomalies. GISTEMP base period 1951-1980.

Video license: CC-BY-4.0
Antti Lipponen (@anttilip)

<https://www.flickr.com/photos/150411108@N06/43350961005/>

Comparisons

How the US compares to other countries in coronavirus tests conducted per million people

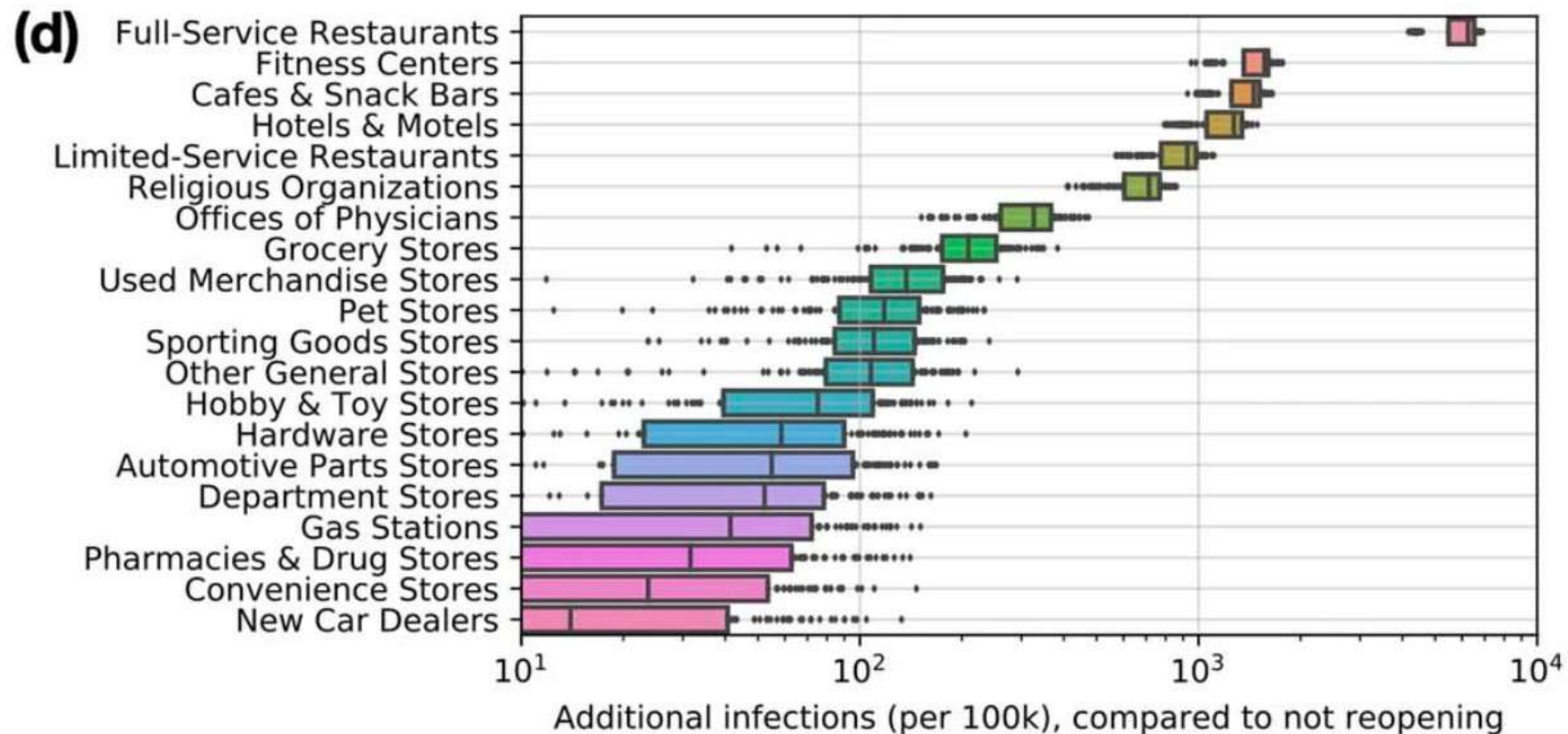


Source: COVID Tracking Project, news reports, government websites; updated April 13 with latest available numbers.

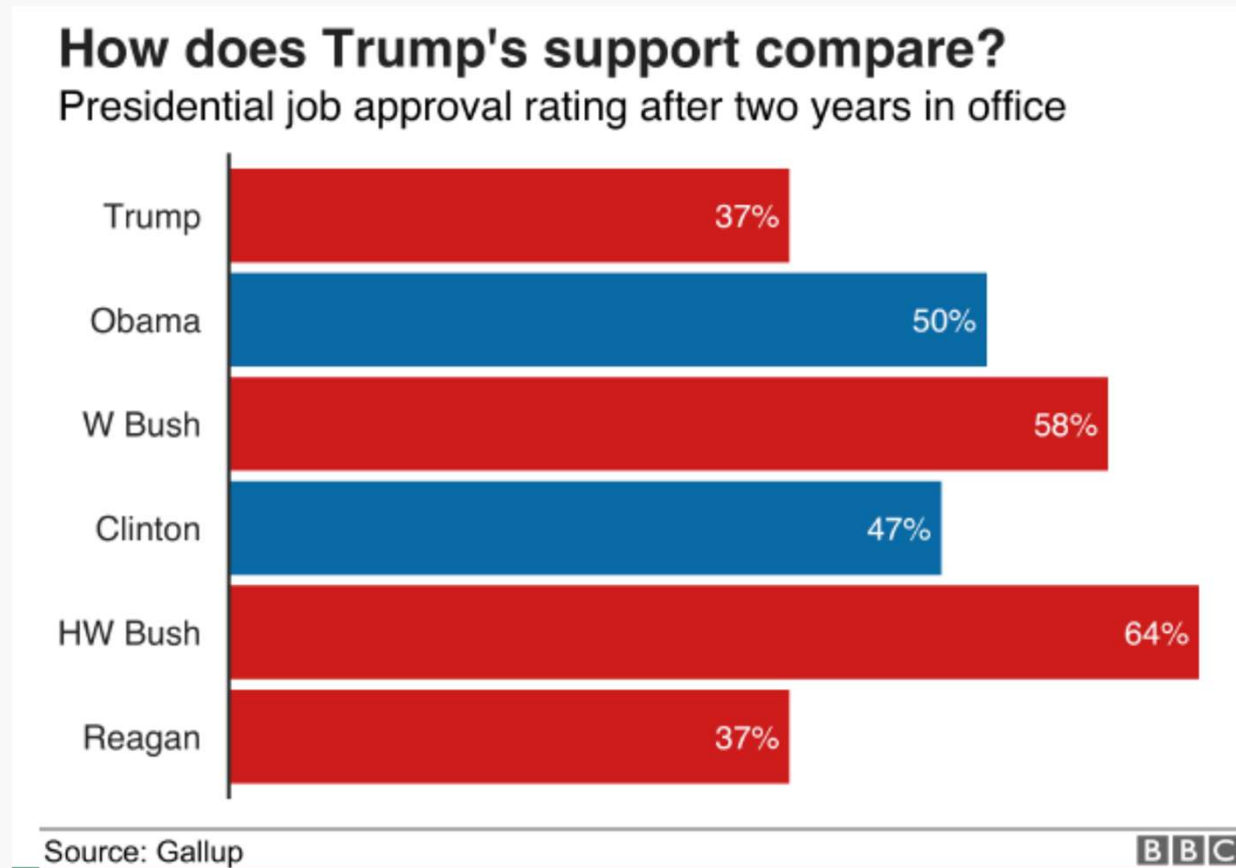
Vox

<https://www.vox.com/2020/4/13/21215133/coronavirus-testing-covid-19-tests-screening>

Box plots COVID infections



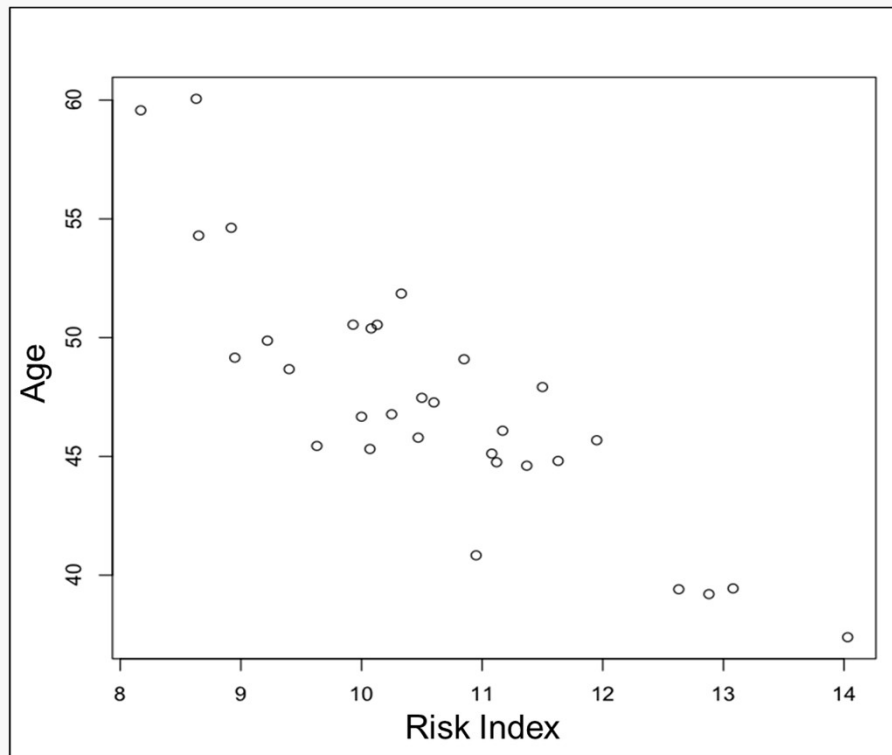
Voting and Political Data



<https://www.bbc.com/news/world-us-canada-38663043>

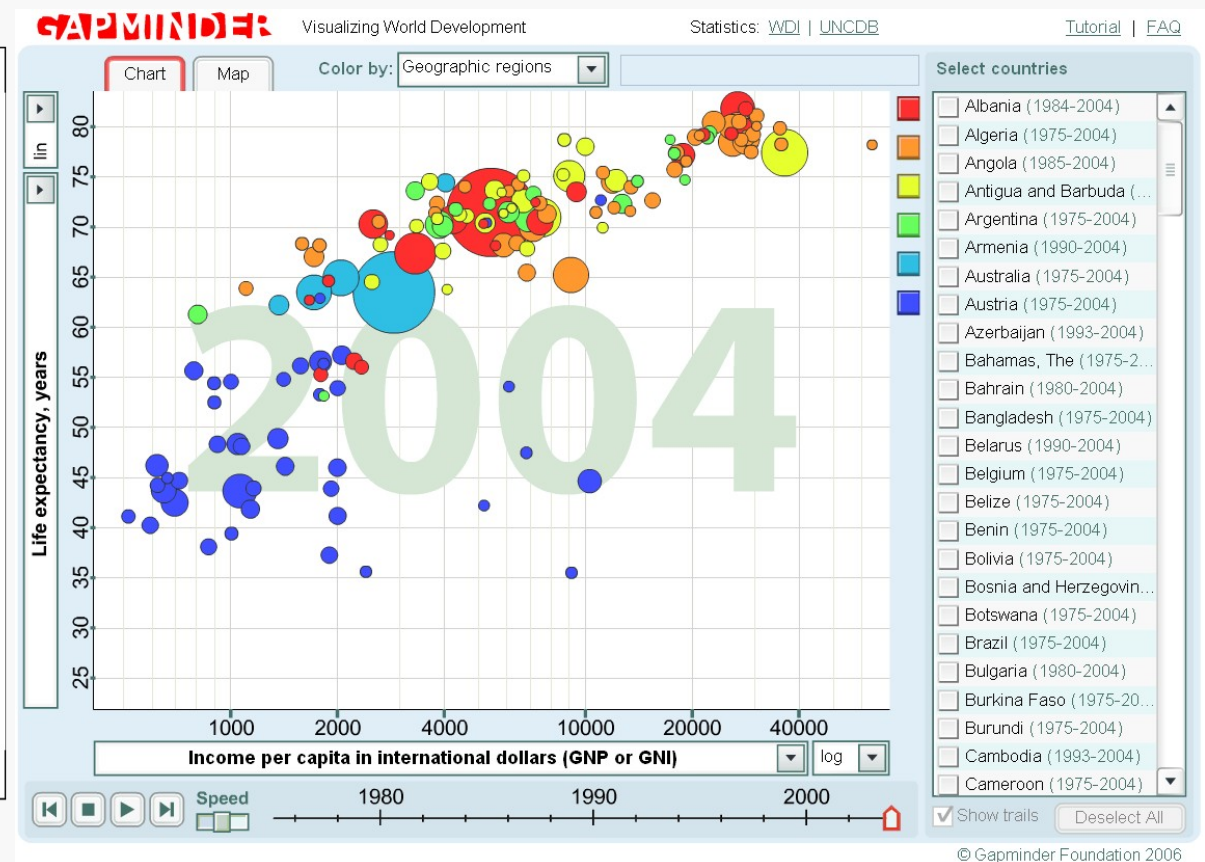
Relationships

Scatter plot



Bubble chart

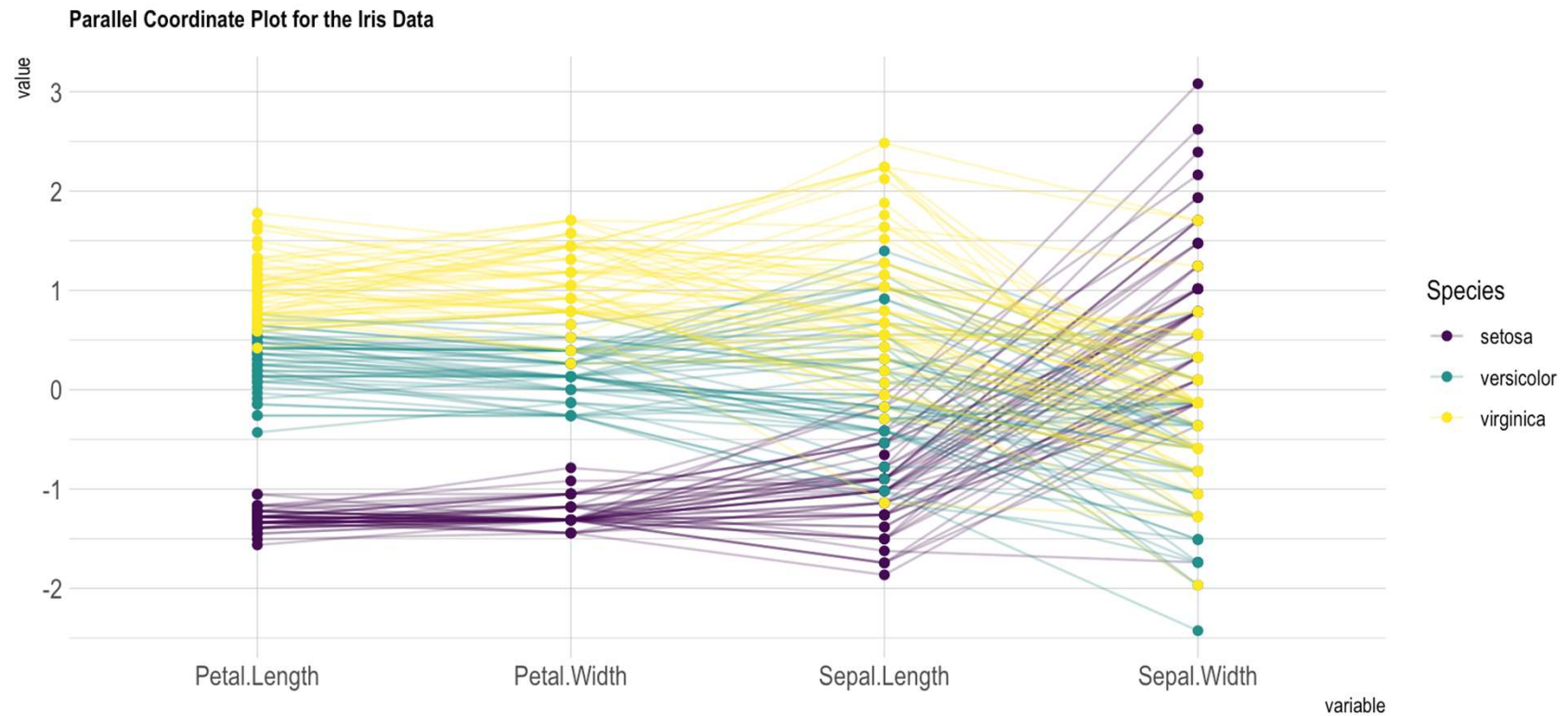
Hans Rosling



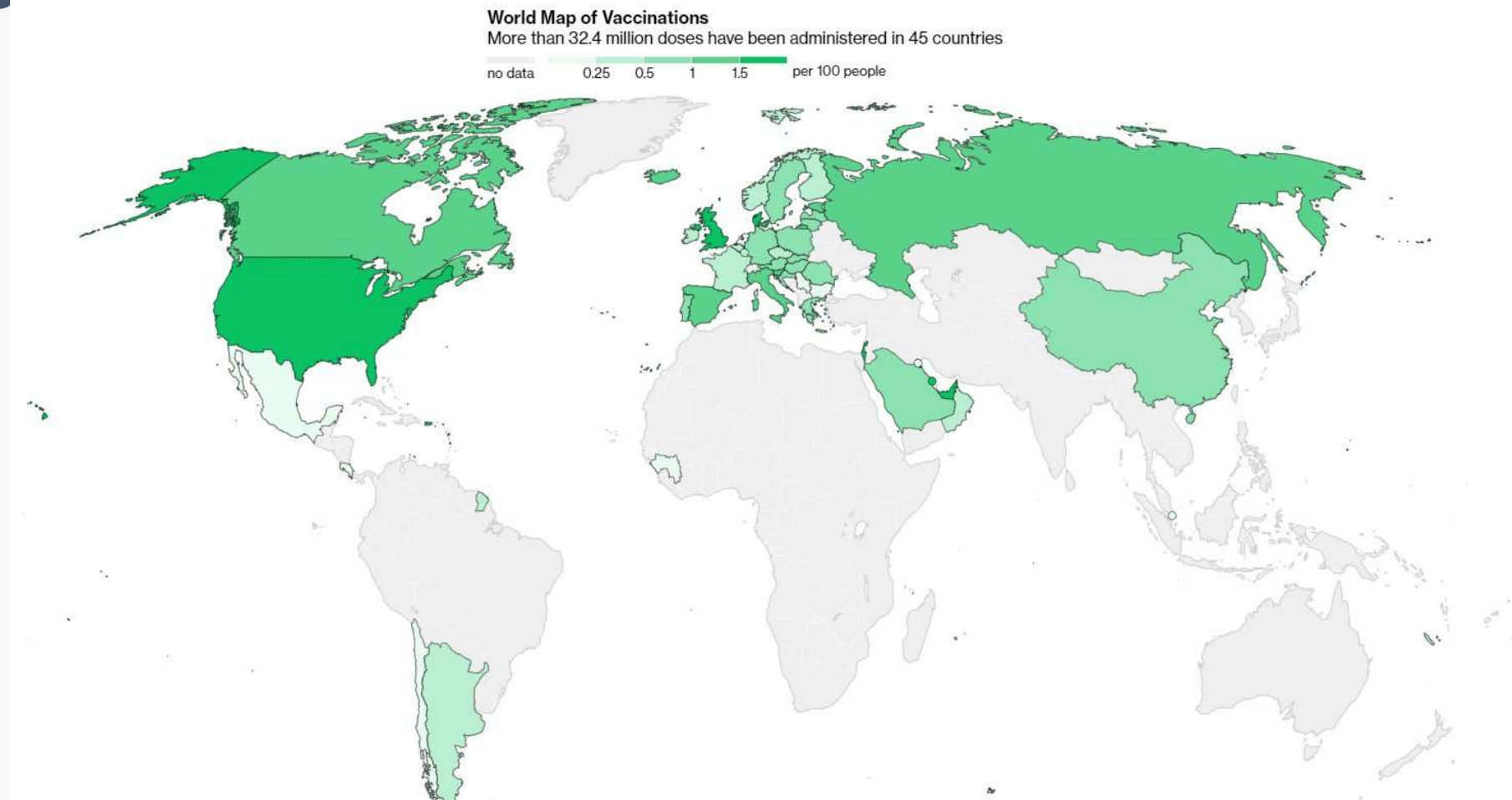
[https://www.ted.com/talks/hans rosling the best stats you ve ever seen](https://www.ted.com/talks/hans_rosling_the_best_stats_you_ve_ever_seen)

[https://www.gapminder.org/tools/#\\$state\\$time\\$value=2020&delay=120;&chart-type=bubbles](https://www.gapminder.org/tools/#$state$time$value=2020&delay=120;&chart-type=bubbles)

Parallel Coordinates

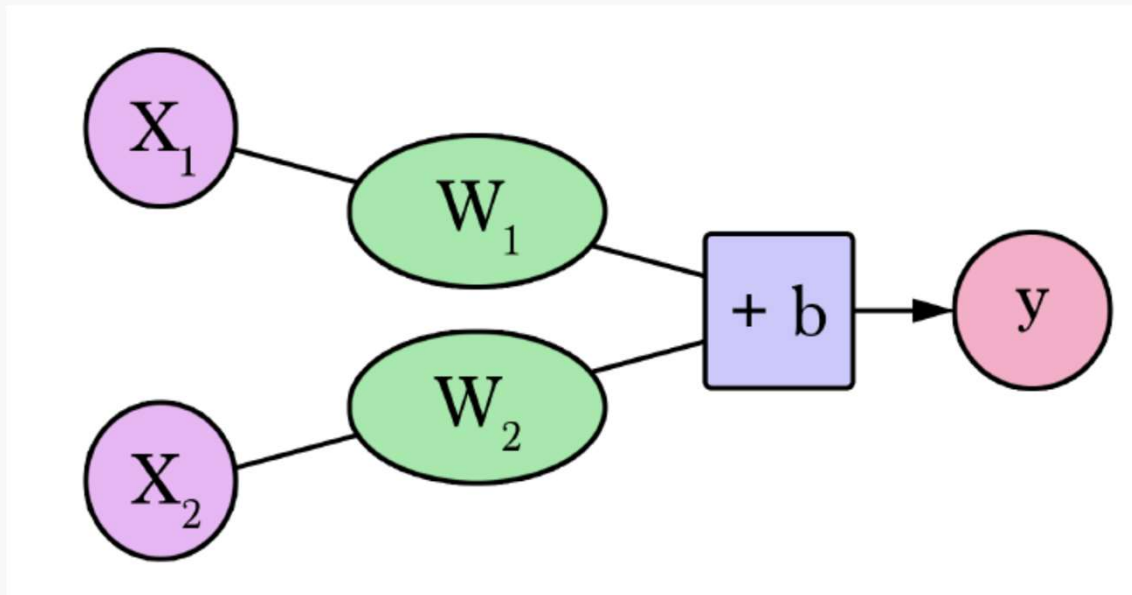


Maps



<https://www.bloomberg.com/graphics/covid-vaccine-tracker-global-distribution/>

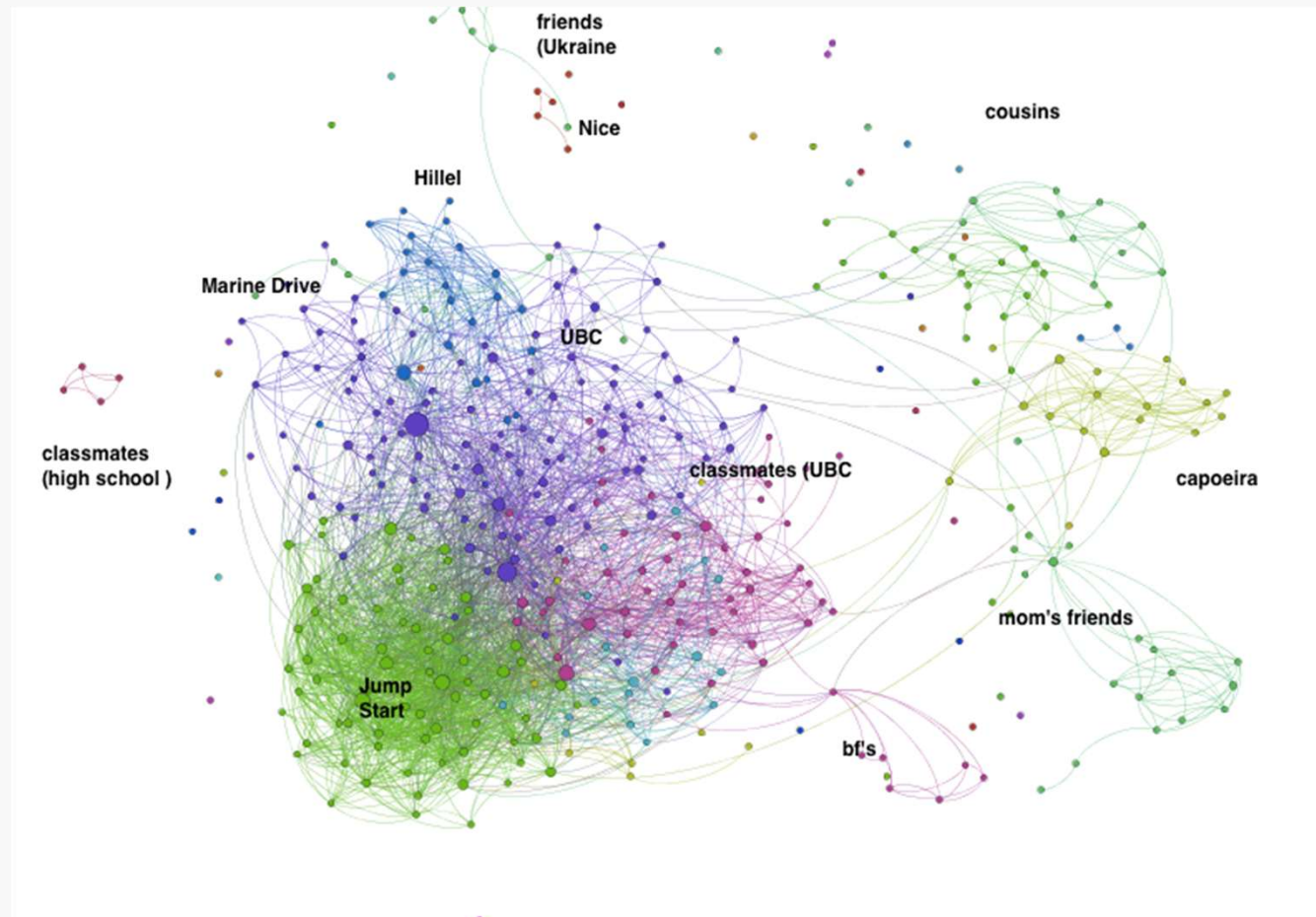
Networks



<http://jalammar.github.io/visual-interactive-guide-basics-neural-networks/>

Networks

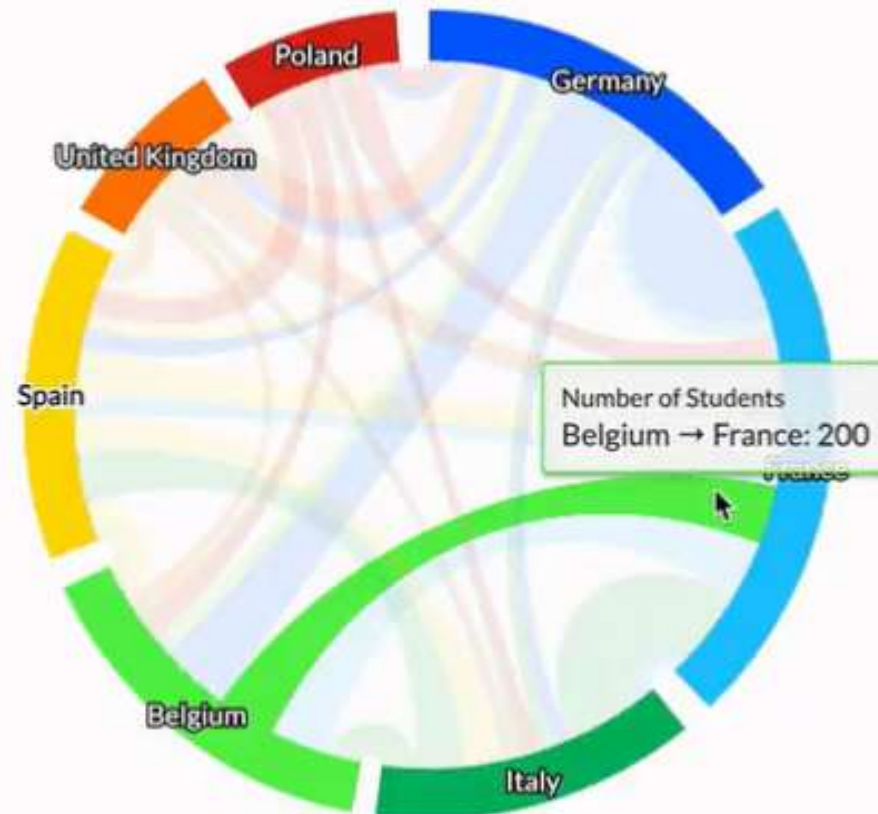
Truthy



[Network Analysis - Social Network Analysis](#)

ERASMUS STUDENT MOBILITY

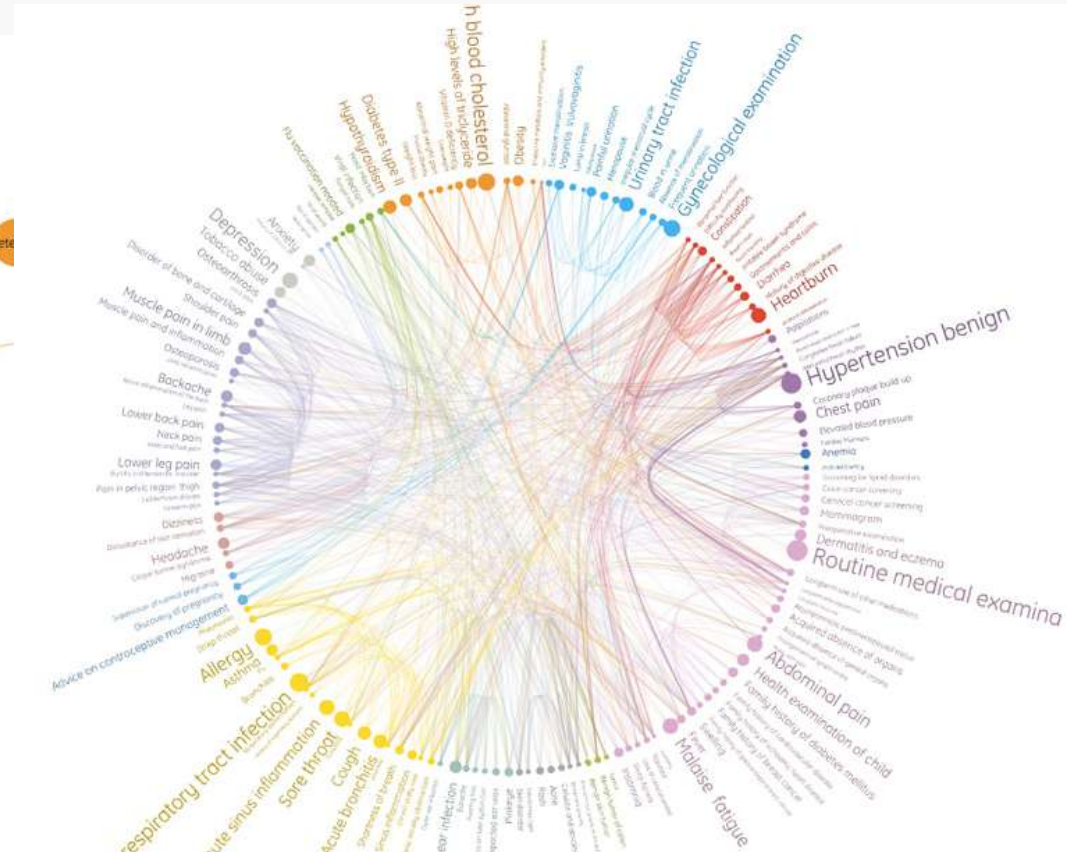
Movement of participants in top participating countries between 2014-2020



Source: Data.Europa

tashapiro shapirotanya

https://twitter.com/tanya_shapiro/status/1617195890627821576?t=av0mPZKrOPHpnSoNIKLqfQ&s=03



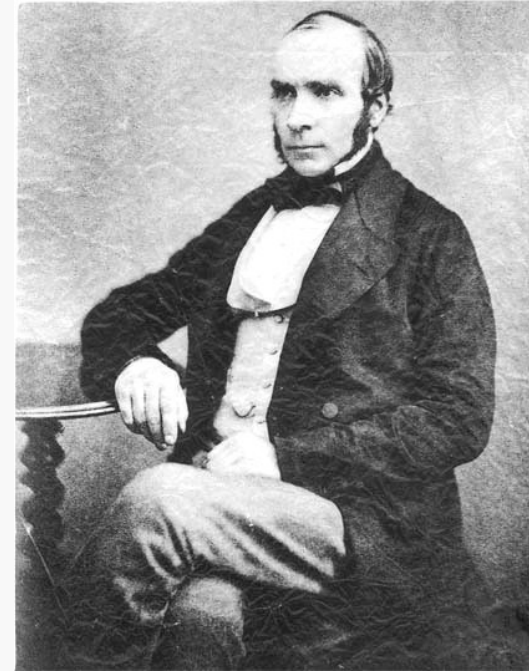
<http://senseable.mit.edu/healthinfoscape/>

1854 London Cholera Epidemic

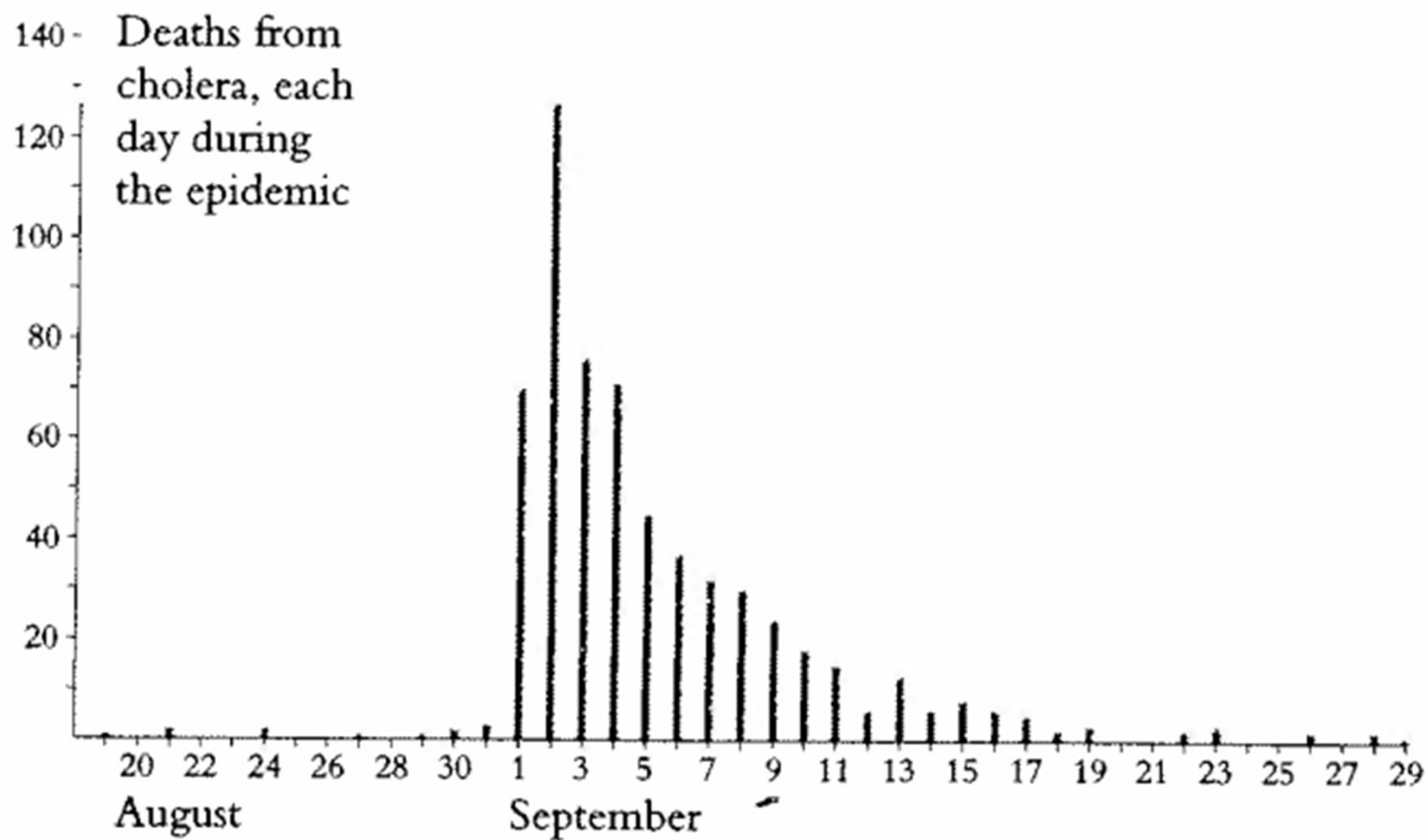
On the evening of August 31, 1854 Cholera broke out in the Broad Street area of central London

John Snow, a London doctor, was asked to investigate

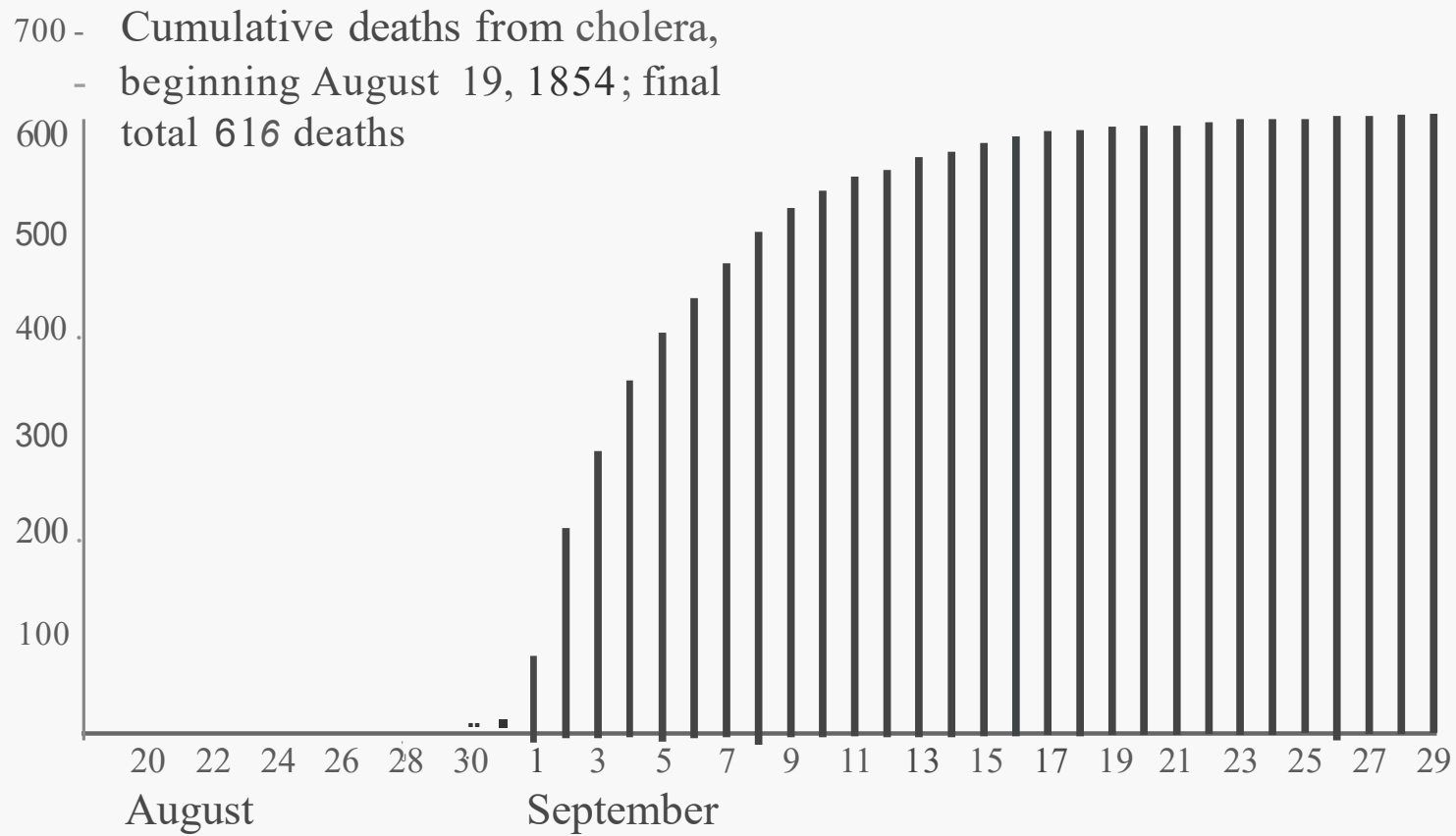
Visualisation was key to his solution



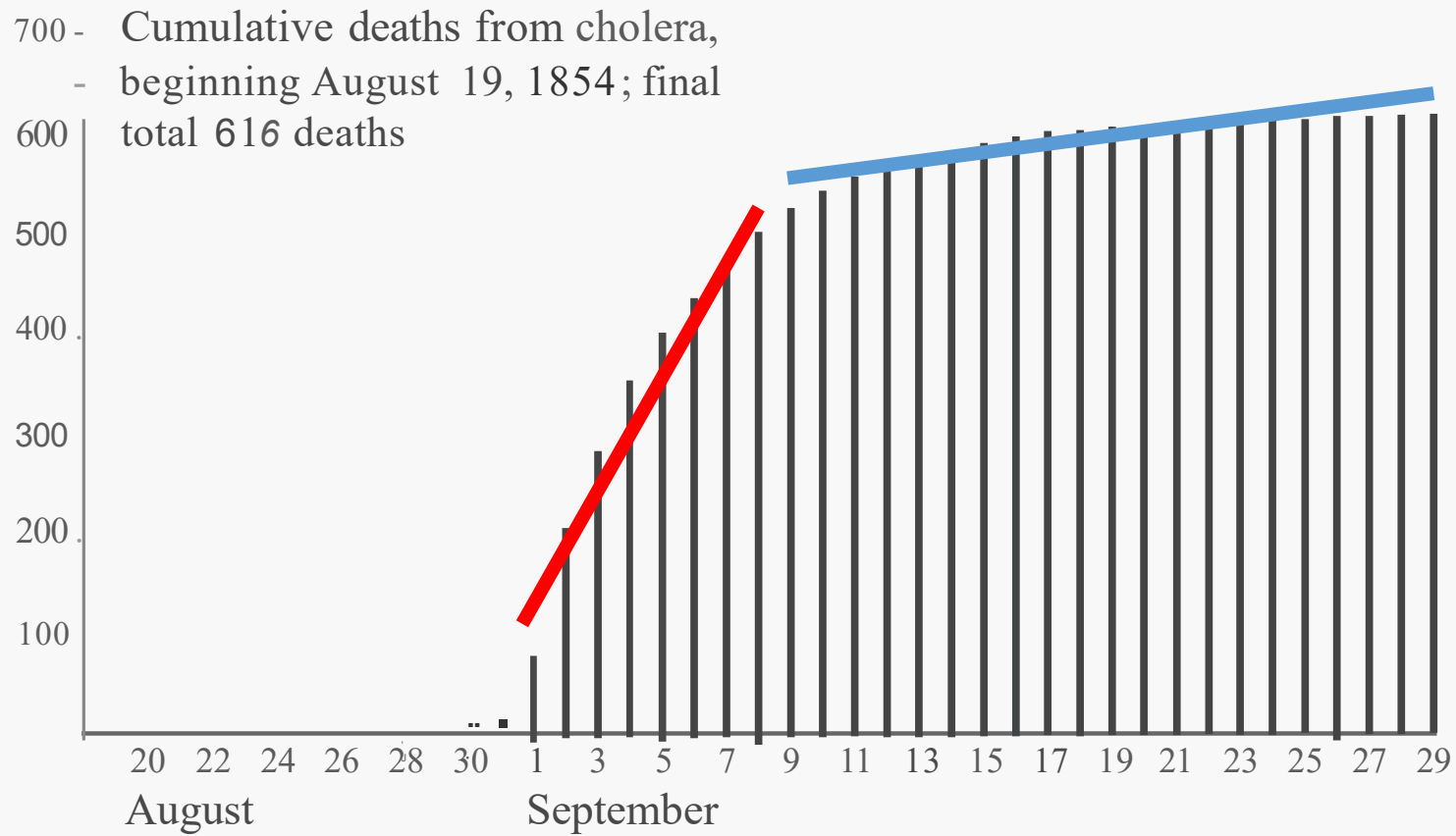
1854 London Cholera Epidemic



1854 London Cholera Epidemic

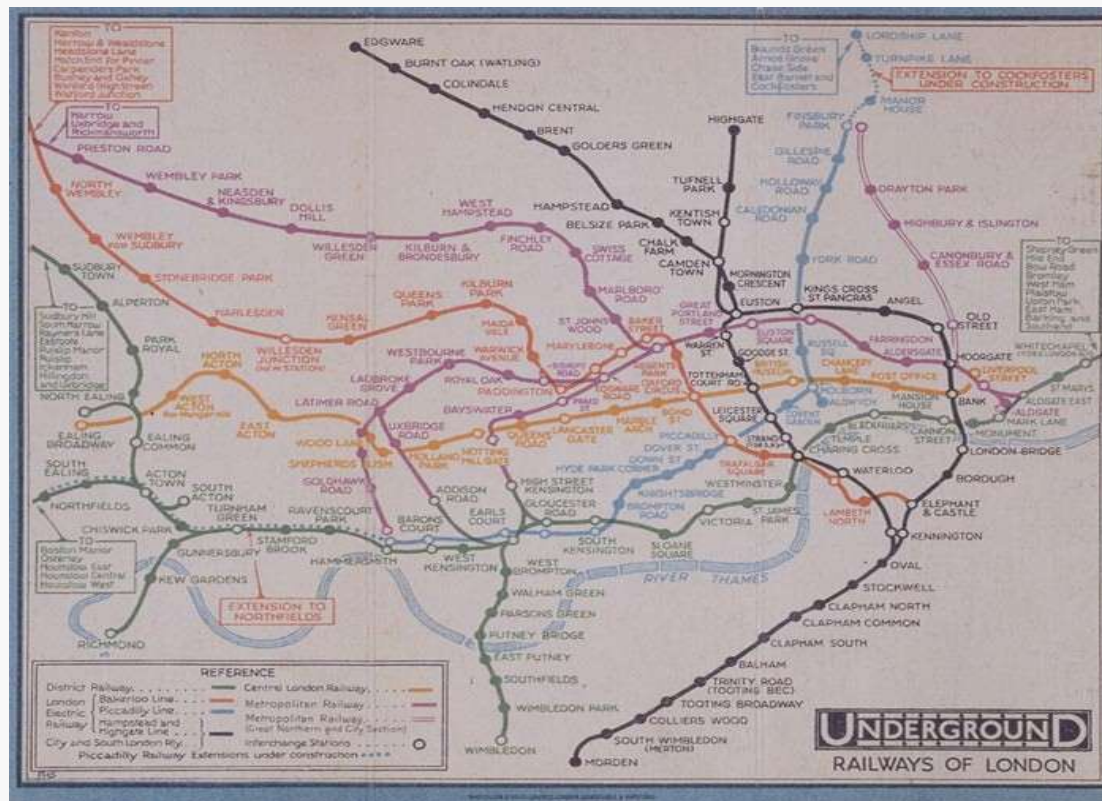


1854 London Cholera Epidemic





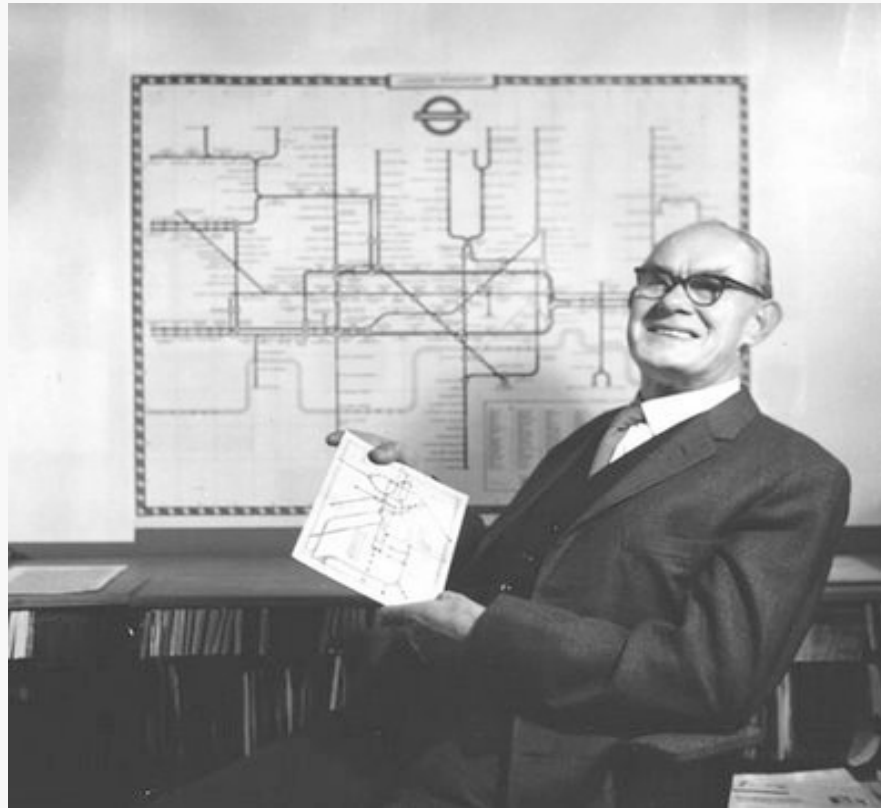
London Underground Map



London Underground map circa 1930

topographicalmapofgrandcanyon.pages.dev/new-hplcp-the-london-underground-map-a-navigational-masterpiece-2-mytqk-pics/

London Underground Map

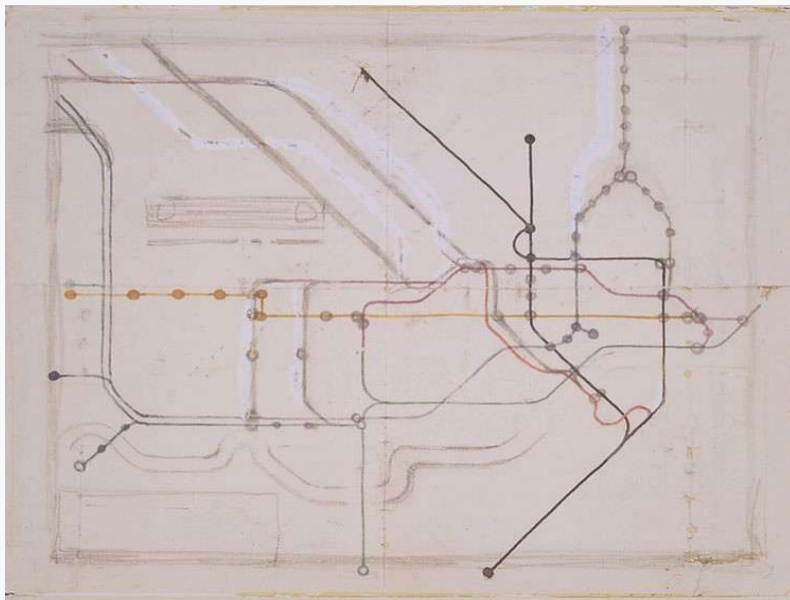


Draughtsman Harry Beck, thought, *"If you're going underground, why do you need bother about geography? ... Connections are the thing"* and designed a new map.

topographicalmapofgrandcanyon.pages.dev/new-hplcp-the-london-underground-map-a-navigational-masterpiece-2-mytqk-pics/

London Underground Map

An early sketch of Harry Beck's new map



Beck's London Underground map circa 1933

topographicalmapofgrandcanyon.pages.dev/new-hplcp-the-london-underground-map-a-navigational-masterpiece-2-mytqk-pics/

Visualisation

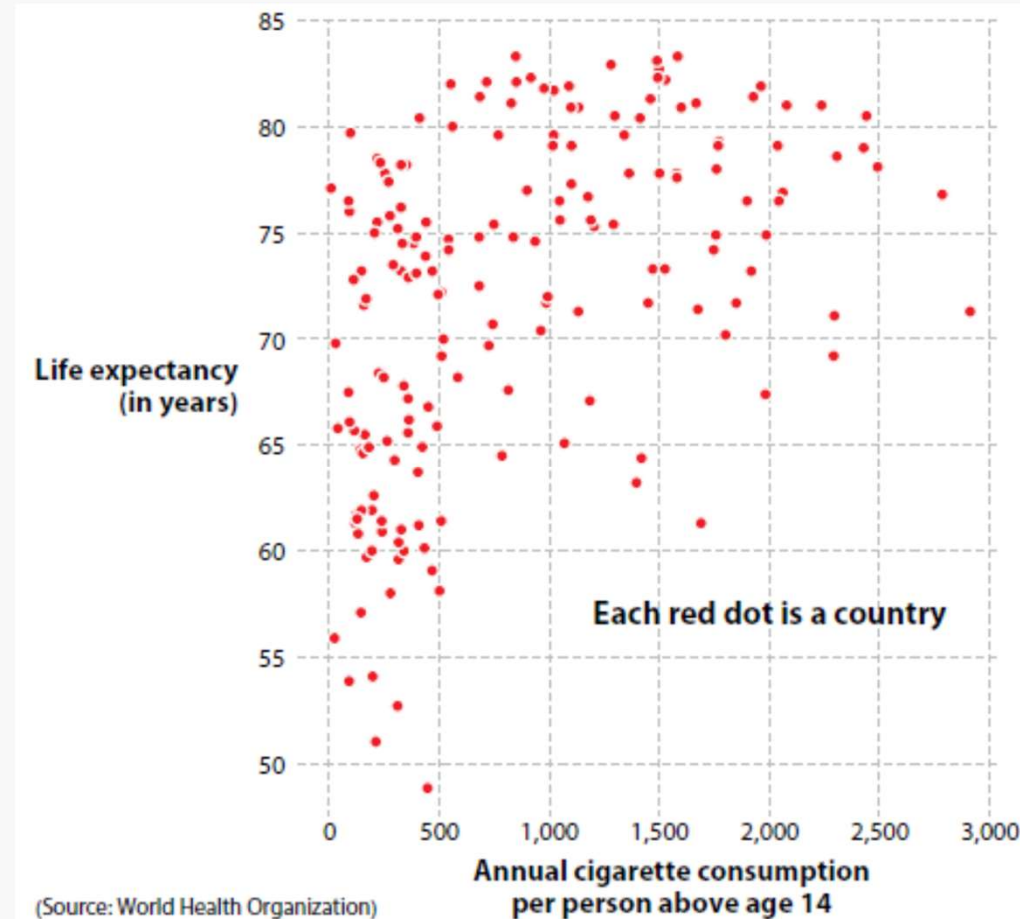
*The challenge is to create effective and engaging visualizations that are appropriate to the data, **the audience and the message***

Charts that Lie

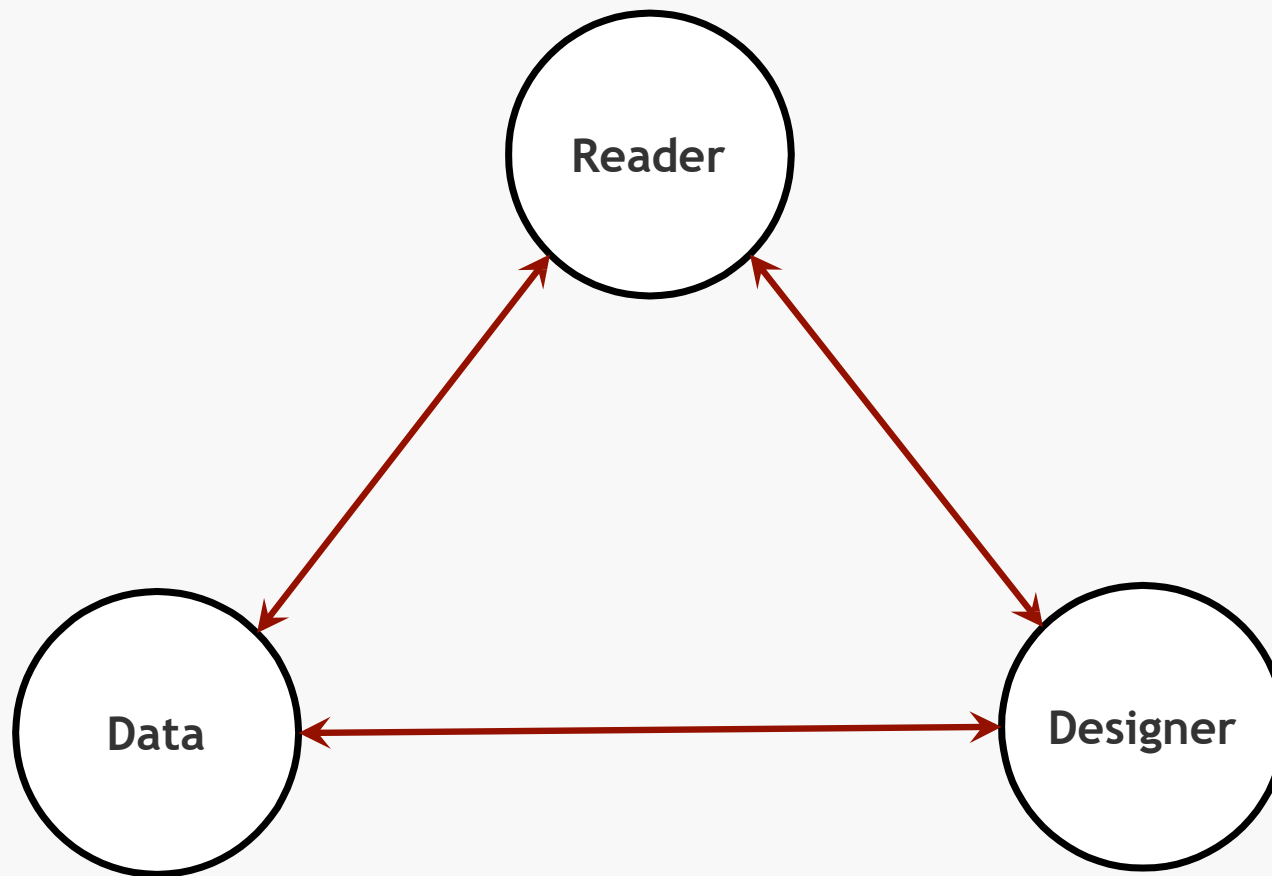
“Good charts are useful because they untangle the complexity of numbers, making them more concrete and tangible.

*However, charts can also lead us to spot **patterns and trends that are dubious, spurious or misleading**, particularly when we pair them with the human brain’s tendency to read too much into what we see and to always try to confirm what we already believe.”*

Alberto Cairo



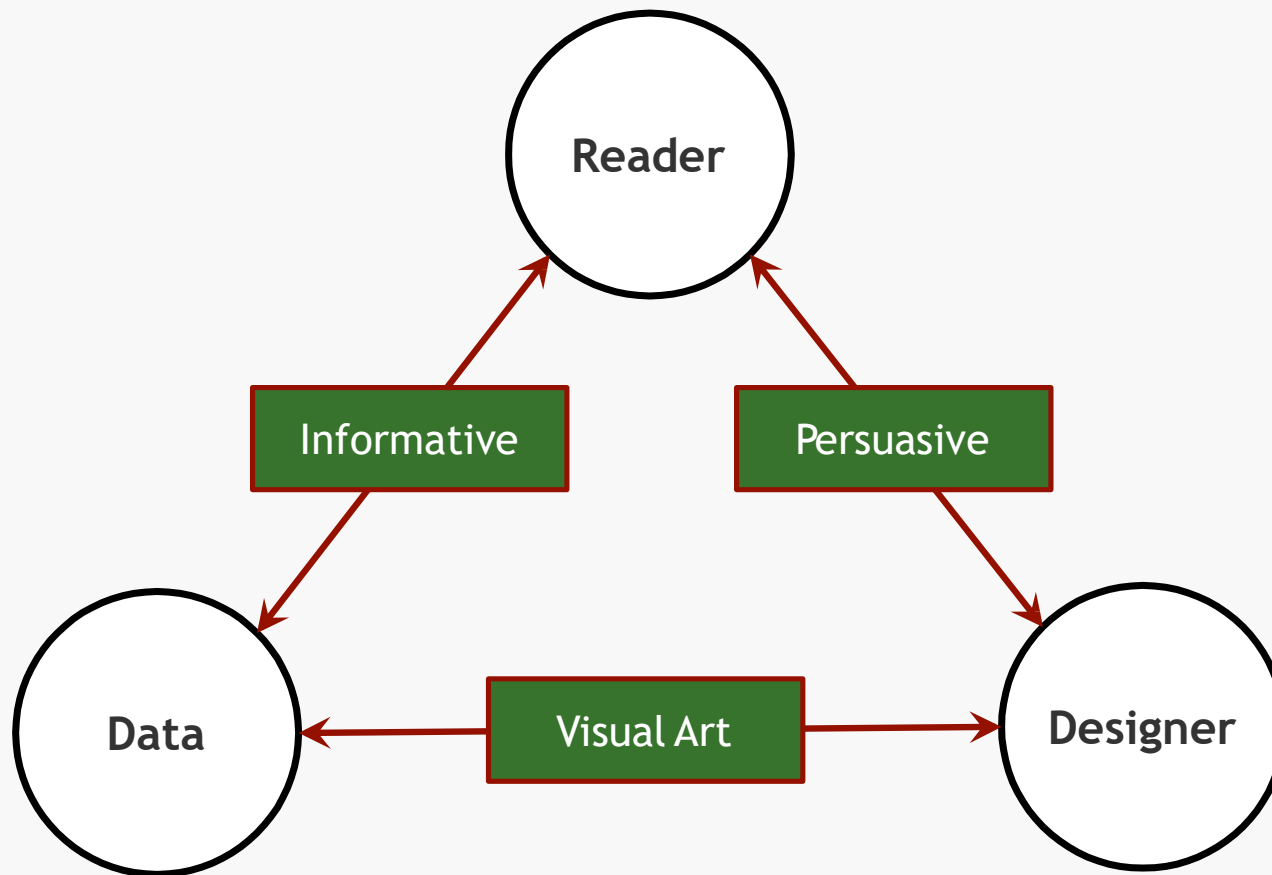
The Visualisation Trinity



“Designing Data Visualizations”, N. Iliinsky & J. Steele, O'Reilly Media, 2011

<http://shop.oreilly.com/product/0636920022060.do>

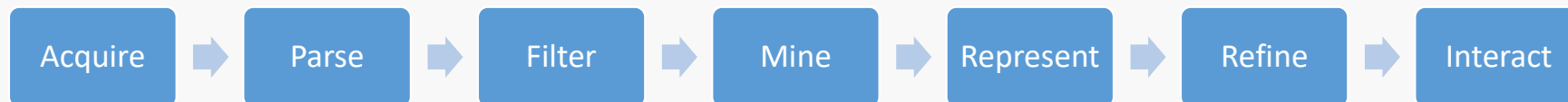
The Visualisation Trinity



"Designing Data Visualizations", N. Iliinsky & J. Steele, O'Reilly Media, 2011

<http://shop.oreilly.com/product/0636920022060.do>

Ben Fry's Visualisation Steps



Acquire

Obtain the data, whether from a file on a disk or a source over a network.

Parse

Provide some structure for the data's meaning, and order it into categories.

Filter

Remove all but the data of interest.

Mine

Apply methods from statistics or data mining as a way to discern patterns or place the data in mathematical context.

Represent

Choose a basic visual model, such as a bar graph, list, or tree.

Refine

Improve the basic representation to make it clearer and more visually engaging.

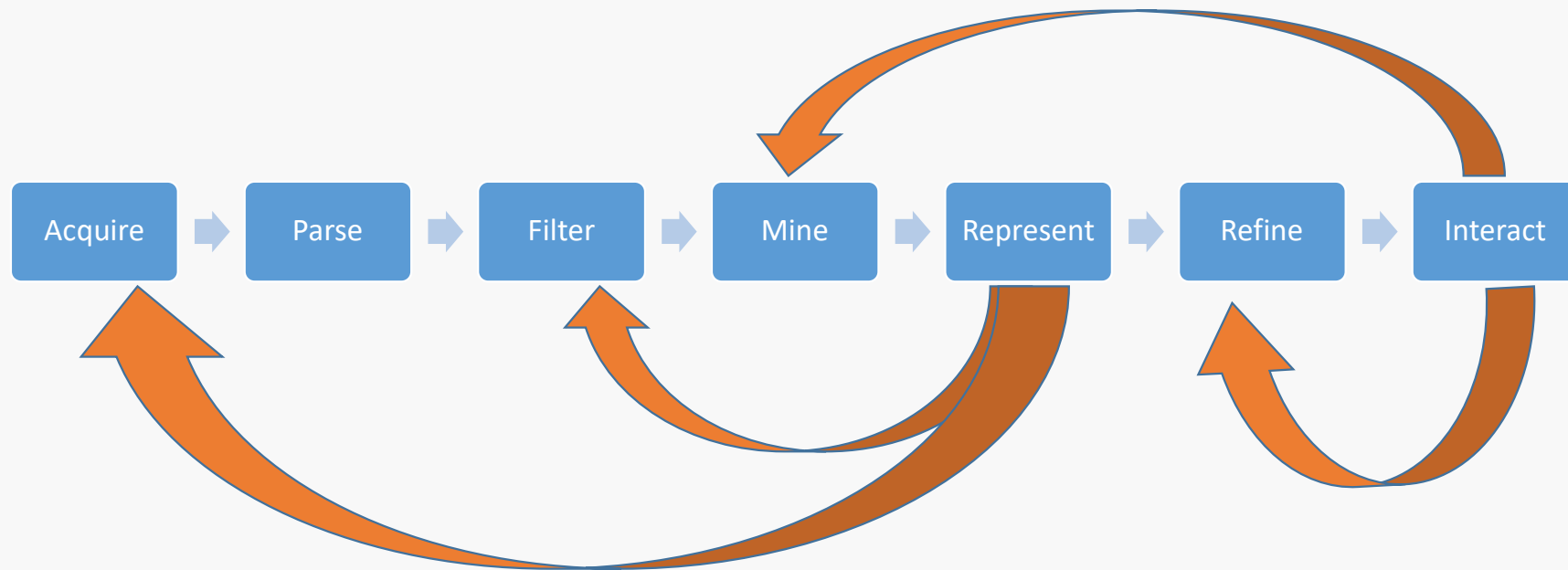
Interact

Add methods for manipulating the data or controlling what features are visible.

"Visualizing Data: Exploring and Explaining Data with the Processing Environment", B. Fry, O'Reilly Media, 2007

<http://shop.oreilly.com/product/9780596514556.do>

Ben Fry's Visualisation Steps



"Visualizing Data: Exploring and Explaining Data with the Processing Environment", B. Fry, O'Reilly Media, 2007

<http://shop.oreilly.com/product/9780596514556.do>

Module Aim

The central aim of this module is help
students attain the ability to

...

**take complex data, process it and
extract value from it through
visualisation**

Learning Outcomes

Create and deploy successful data visualisations using leading software tools

Design effective visualizations based on principles from perceptual psychology, cognitive science, graphic design and visual art

Select, formulate and integrate metaphors to suit particular data-driven tasks

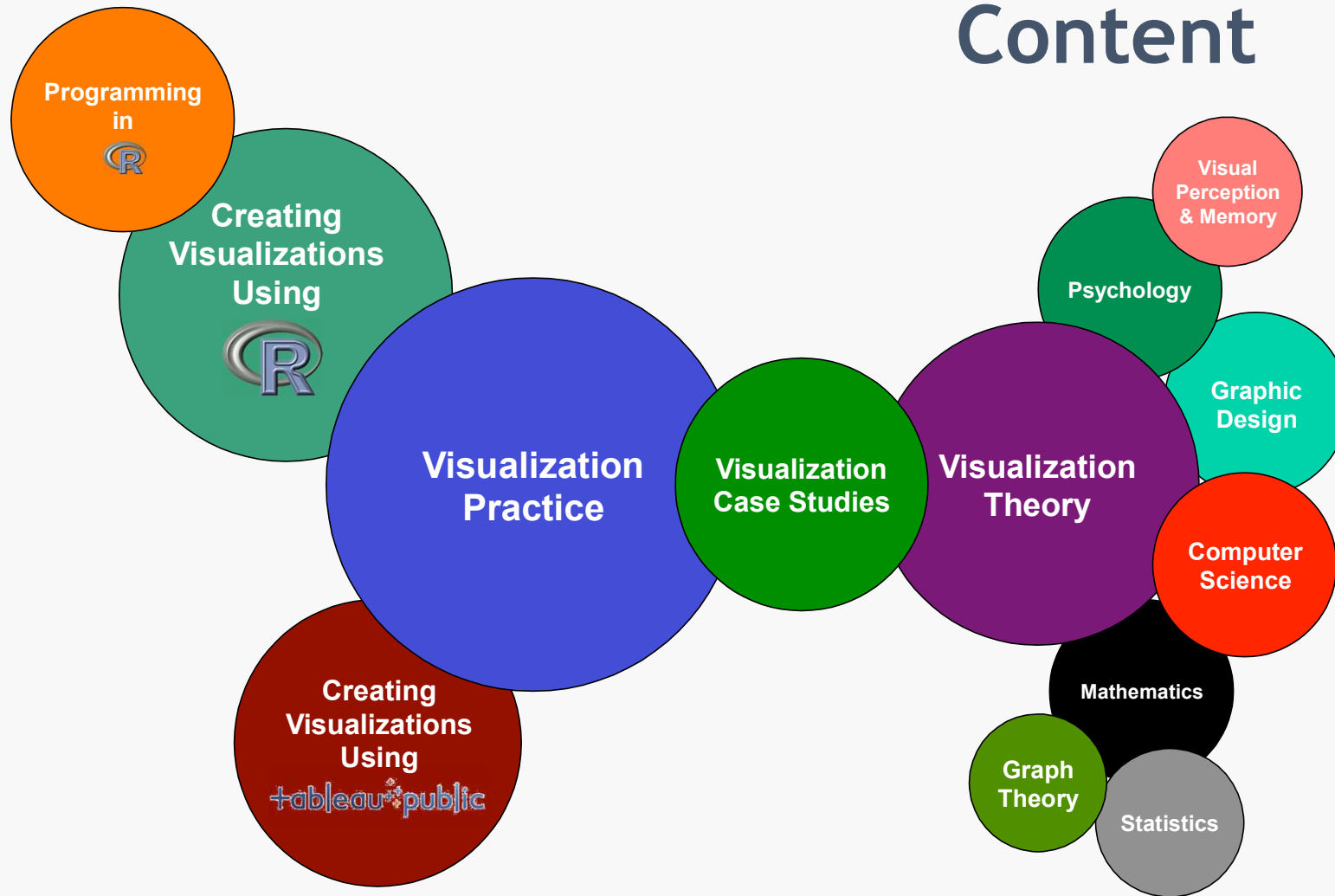
Analyse and evaluate how mental models aid in the interpretation of complex visual displays

Demonstrate understanding of how humans perceive the world around them on a general level and absorb complex data/information on a specific level

2011/12

Analyse and evaluate how metaphors are used to convey unfamiliar information

Content



Module Content

Fundamentals of visualisation

Data characteristics
& dimensions

Use of encodings

Colour theory

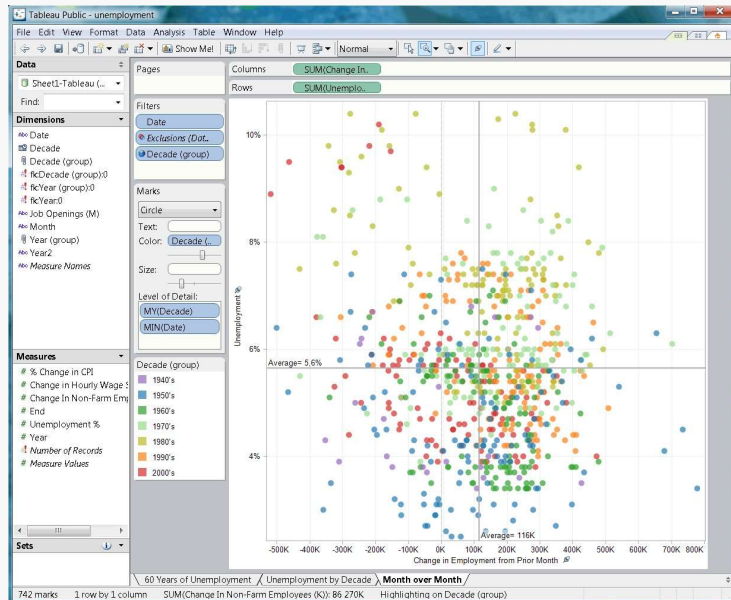
Graphical perception
& communication

Interaction design

Visualisation types:

- Trends
- Comparisons
- Correlations
- Maps
- Networks
- Hierarchies
- Visualising Text
- High-dimensional data

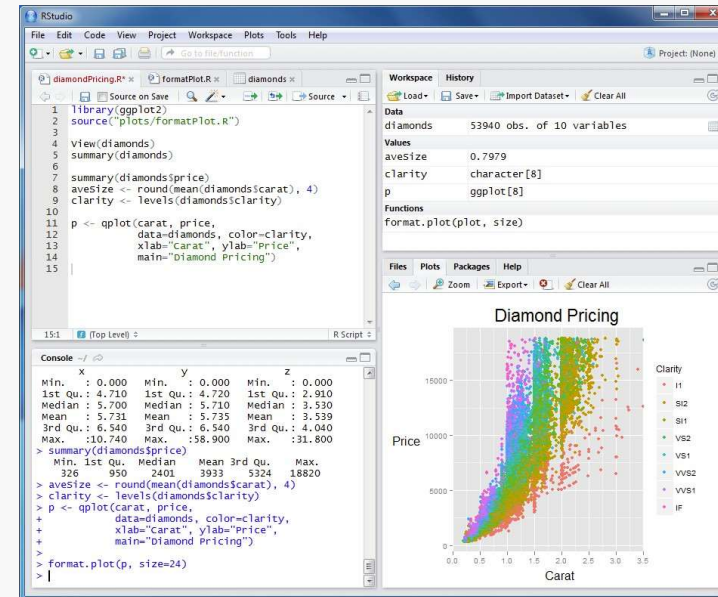
Tools



+tableau++public

Tableau is an enterprise level visualisation-based business analytics tool

<https://www.tableau.com/products/public>



R & R

R is a statistical programming language with extensive visualisation features

R Studio is a rich GUI interface to R

www.r-project.org

[RStudio Desktop - Posit](#)

Tableau Public Visualisations



HOW IT WORKS
see the magic happen

GALLERY
see what others visualized

COMMUNITY
join the conversation

See what's possible with Tableau Public interactive visualizations.



Tale of 100

Author: Christian Chabot published on [Wall Street Journal](#)

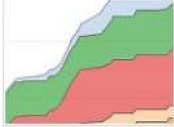
How fast do successful tech companies grow? The Wall Street Journal posted this visualization that compares the performance of 100 fast growing software companies.



College Football Recruiting

Author: Scott Wasserman published on [Tableau Software](#)

This year's national championship might have just been decided, but the struggle for years of future domination has already started. The recruiting wars are in full swing. Use this viz to see how your team stacks up.



Tech IPOs Finish the Year Strong

Author: Daniel Horn published on [IPO Dashboards](#)

Tech IPOs have been under scrutiny this year but ended up leading the way in total capital raised during 2011.



Making the Pay Gap Look Good

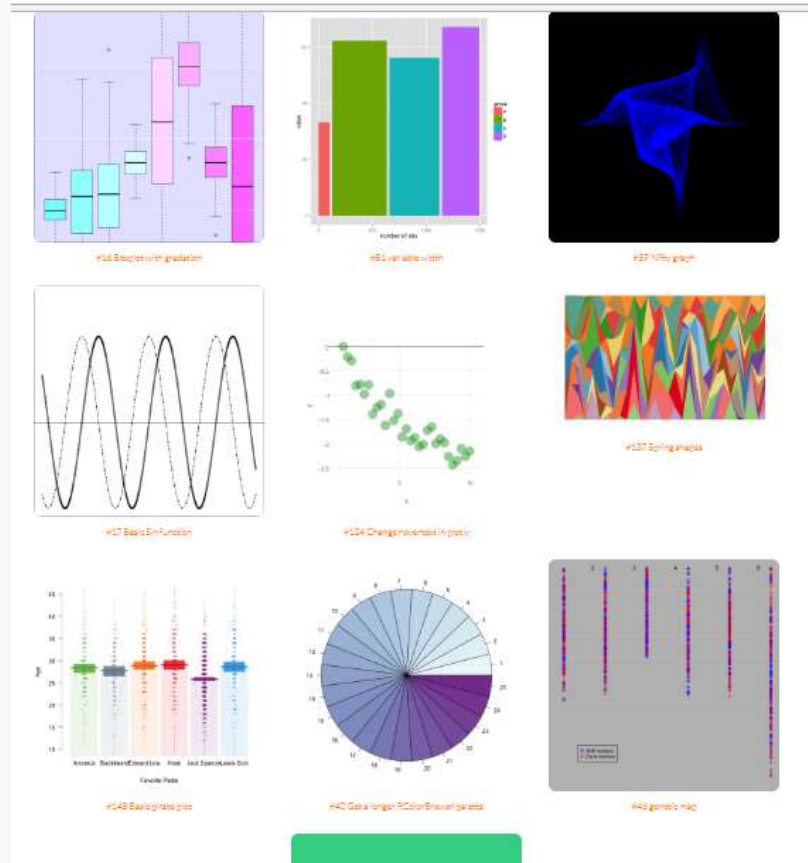
Author: Simon Rogers published on [The Guardian Datablog](#)

Simon Rogers makes the gender gap in salaries look like a million in this viz. His use of a gantt bar to

A gallery of visualisations created using Tableau Public

www.tableausoftware.com/public/gallery

R Visualisations



A gallery of visualisations
created using R with
source code!

<https://www.r-graph-gallery.com/all-graphs/>

Books

1. Designing Data Visualisations – Noah Iliinsky, Julie Steele (2011)
2. Visualise this – Nathan Yau (2011)
3. Visualising Data – Ben Fry (2008)

Also worth looking up:

4. The Visual Display of Quantitative Information – Ed Tufte (1983)
5. Semiology of graphics – Jacques Bertin (1967)
6. Andy Kirk – Data Visualisation (2016)
7. How Charts Lie – Alberto Cairo (2019)

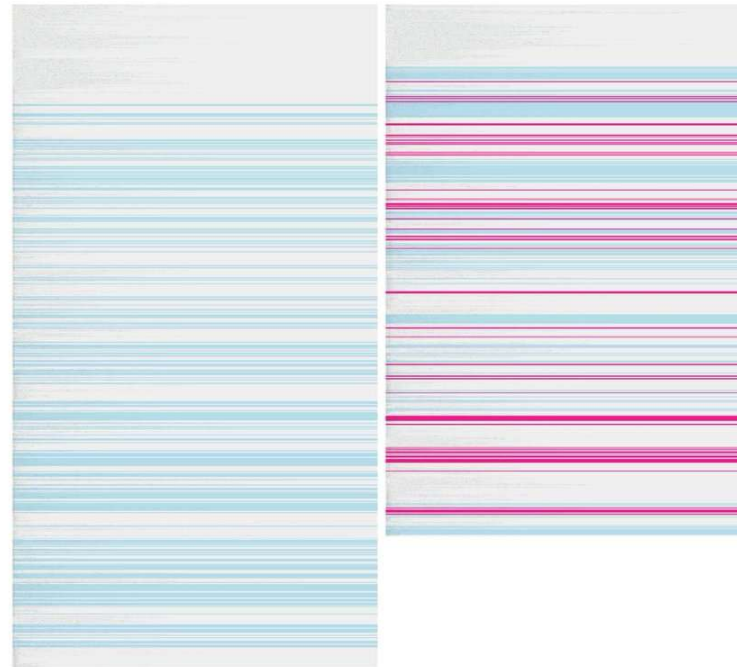
Visualization of the Week

Every time Ford and Kavanaugh did or did not answer a question

■ Answered question ■ Did not answer

FORD

KAVANAUGH

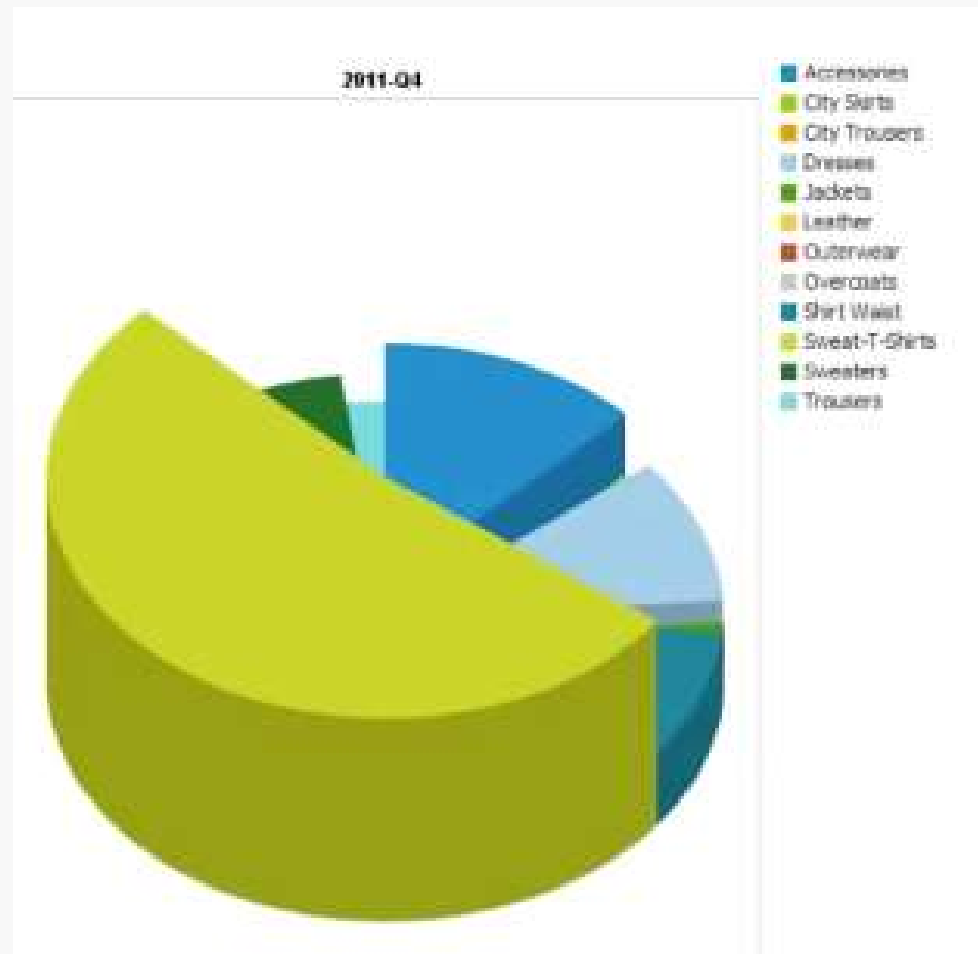


Data from a Vox analysis of a Bloomberg transcript of the hearing,
via the Washington Post

Vox

<https://www.vox.com/policy-and-politics/2018/9/28/17914308/kavanaugh-ford-question-dodge-hearing-chart>

(Un)Visualization of the Week



The Joy of Stats

Hans Rosling

<https://youtu.be/jbkSRLYSojo>

Thanks To

John McAuley, Colman McMahon and
Brian Mac Namee
for an earlier versions of these lecture notes